



Study The Propagation Of Computer Virus In Network Environment

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CERTIFICATE

This is to certify that Mr. K.Gowtham Sricharan, pursuing B.Tech course at Indian Institute of Technology, Kharagpur in Computer Science and Engineering has undertaken a project as an intern at IDRBT, Hyderabad from May 5, 2014 to July 5, 2014.

He was assigned the project "Study The propagation of Computer Virus in Network Environment" under my guidance. During the course of the project he has undertaken a study of Mathematical Modelling and Security in Computer Networks and also has done excellent work.

I wish him all the best for all his endeavours

Dr. N Raghu Kisore
(Project Guide)
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I am also thankful to Indian Institute of Technology, Kharagpur for giving me this golden opportunity to work in a high-end research institute like IDRBT

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1.Introduction

Propagation of computer virus is a great threat to the Internet security. On July 2001, the worm CodeRed infected more than 359,000 hosts on the Internet within only 14 hours .The experience by explosive propagation of CodeRed makes both network administrators and users aware of the importance of counteraction against the computer virus. Recently, another type of malware, Botnet is widely spread through the Internet. Likewise the traditional computer virus, Botnet propagates themselves to vulnerable Internet host

In the Internet society, the counteraction against computer virus propagation plays an important role to build highly dependable and secure system. A virus scan by the anti-virus software is the most basic but powerful counteraction. The anti-virus software scans all the files in the system to find known virus patterns, and gets rid of or quarantines the virus infected files. In general, the virus pattern file for scanning is distributed by the vendor of anti-virus software. However, the pattern files cannot always be ready for the distribution just after a new computer virus is discovered. Thus the attack may occur before receiving the patternfile for the new virus, i.e., zero-day attack. Removing the threat of zero-day attack is still a significant challenge in computer security. Statistical approach is a promising approach to discover unknown computer viruses from the Internet traffic, and can protect us from the zero-day attack. In fact, several organizations and security companies are monitoring the Internet traffic to detect abnormal traffic caused by computer virus

Zou et al. proposed an early detection method for the computer virus propagation. Their approach was based on a parametric epidemic model and its related statistical estimation. In general, the epidemic models were often applied to measuring and characterizing the computer virus propagation. Murray seems to be the first work to formulate the relationship between epidemiology and propagation of computer virus. Kephart and White studied prevalence of computer virus in the context of deterministic epidemic model, called susceptible

– infectious – susceptible (SIS) and susceptible – infectious – recovered (SIR) models. In particular, they also introduced the effect of a warning called kill signal (KS) in the framework of their SIR model. The KS is regarded as a kind of rumors for the computer virus. Unlike usual immunity in biology, the KS rapidly spreads over the Internet with comparable speed to the worm propagation. Sellke et al. also proposed a stochastic model for the computer virus based on epidemiology models.

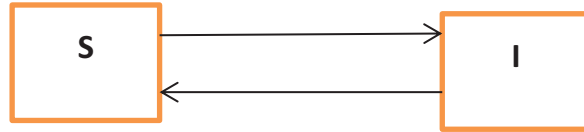
Computer virus propagation is influenced by various factors, and these factors are regarded as constants in most of the existed models. So, some detail information of computer virus propagation is neglected, and mathematical model is simplified. In fact, many factors are changed during the virus propagation. In this paper, the virus propagation rate is designed as a variable for simulating exactly

2. Deterministic Models

Various models have been used in the field of epidemiology. SIS and SIR models are the most prominent ones. These models are deterministic models and are valid only in case of sufficiently large populations. The transition rates from one class to another are mathematically expressed as derivatives, hence the model is formulated using differential equations. While building such models, it must be assumed that the population size in a compartment is differentiable with respect to time and that the epidemic process is deterministic. In SIS and SIR epidemic models, individuals in the population are classified according to disease status, either susceptible, infectious, or immune. The immune classification is also referred to as removed because individuals are no longer spreading the disease when they are removed or isolated from the infection process. These three classifications are denoted by the variables S , I , and R , respectively

2.1 SIS model

By using epidemic theory we can model worm propagation. In an SIS model, a susceptible node, after a successful infection by the worm propagation becomes infected and infectious, but does not develop immunity to the worm. Hence, after recovery, infected host return to the susceptible class i.e still in a vulnerable state



Let $s(t)$ and $v(t)$ denote the deterministic numbers of vulnerable and worm-infected hosts, respectively. Then the dynamics on the numbers of vulnerable and worm-infected hosts in the network can be described by

$$\frac{dv(t)}{dt} = \beta s(t)v(t) - \delta v(t)$$

$$\frac{ds(t)}{dt} = -\beta s(t)v(t) + \delta v(t)$$

where $\beta (> 0)$ and $\delta (> 0)$ are the infection rate and removal rate respectively and $K=S(t)+I(t)$ is the total population size. The initial conditions satisfy $S(0) > 0, I(0) > 0$, and $S(0) + I(0) = K$. The dynamics of model are well-known . They are determined by the basic reproduction number. The basic reproduction number is the number of secondary infections caused by one infected host in an entirely susceptible population. For model , the basic reproduction number is defined as follows

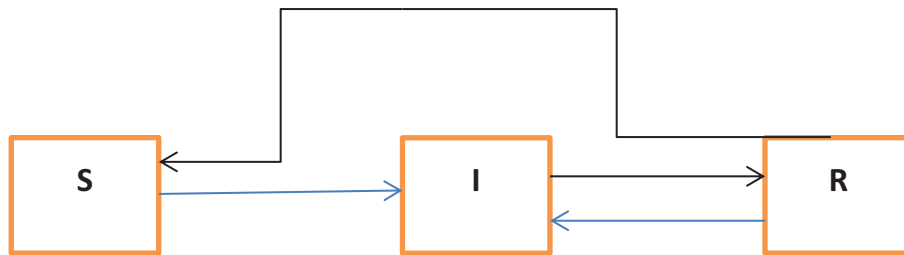
$$R_o = \frac{\beta}{\delta}$$

The number of infections follows the logistic model

$$\frac{dv(t)}{dt} = \{\delta(R_o - 1) - \beta v(t)\}v(t)$$

2.2 KS(Kill Signal) Model

The SIS model is a simple epidemic model to characterize the propagation of computer virus, but cannot take account of the immunity to computer virus. The model with the framework of immunity is often called the susceptibleinfected removed/immune-susceptible(SIRS) model. In the computer virus prevalence, Kephart and White introduce the concept of kill signal(KS), which is regarded as a warning for propagation of computer virus.



Let $v(t)$ and $w(t)$ denote the number of infected and non-vulnerable hosts at time t , respectively. Then the dynamics of the infected and nonvulnerable hosts are described by the following differential equations:

$$\frac{d}{dt}v(t) = \{K - \delta - \beta_r + \beta_r\}w(t) - \beta v(t)\}v(t)$$

$$\frac{d}{dt}w(t) = \{K - \delta_r - \beta_r\}w(t) - \beta w(t)\}w(t) + \delta v(t)$$

where K is the total number of hosts in the network, β is the propagation rate of computer virus, δ is the removal rate of computer virus, β_r is the KS-spreading rate and δ_r is the re-vulnerability rate.

3. Stochastic Models

But in case the population is small or when parameters like the infection rates are transient in nature stochastic models are preferable. Stochastic epidemic models are described by birth and death processes which have specific state transitions. Dissimilar to deterministic models, stochastic models describe the probabilistic behaviour of virus propagation using probability mass functions for Markov states, and can represent rare events such as virus extinction. These models have an absorption state where number of infected nodes is zero. That is they assume that a virus attack eventually gets terminated with a probability of 1. Most of the published work build stochastic SIS and SIR models based on Markovian arrival process (MAP). MAP is a counting process whose arrival rate is governed by a Continuous Time Markov Chain (CTMC). The number of infected hosts is represented by a Continuous Time Markov chain (CTMC). Some of the measurements made by a stochastic model are:

- the basic reproduction number
- the probability of virus extinction
- the mean time to virus hazard
- the mean time to virus extinction

In each of the above cases the most important metric that defines the ability of an infection to spread is the reproduction number, R . The following inferences can be made from value of reproduction number, R

- ≤ 1 indicates the whole process converges to disease free state.
- > 1 indicates the infection to be epidemic in nature and infection would spread until there is no more susceptible population.

3.1 Stochastic SIS model

In the stochastic modelling, dynamics of propagation can be modelled by a birth-and-death process. Especially, the stochastic SIS model is introduced from the classical epidemic theory to analyze worm propagation.

If there are K computer hosts in the network, the worm propagation is given by the birth-and-death process with the following birth and death rates:

$$\lambda_i = \beta(K - i)i, \quad i = 0, 1, 2, \dots, K$$

$$\mu_i = \delta i, \quad i = 0, 1, 2, \dots, K$$

where $\beta(> 0)$ and $\delta(> 0)$ are the propagation rate of worm and the removal rate of infected worm respectively. Note that the birth rate λ_i depends on the number of both worm-infected and vulnerable hosts in the network. Also in the stochastic SIS model, it is assumed that the host from which a worm is removed is still vulnerable.

Let a stochastic process $\{V(t) : t \geq 0\}$ be the number of worm-infected hosts at time t , and $p_v(t), v = 0, \dots, K$ denote the probabilities that there are v worm-infected hosts in the network at time t . Based on the analysis technique of CTMC, the probabilities are given by the following

$$\frac{1}{dt}p_0(t) = \mu_1p_1(t)$$

$$\frac{1}{dt}p_1(t) = \mu_2p_2(t) - (\lambda_1 + \mu_1)p_1(t)$$

$$\frac{1}{dt}p_i(t) = \lambda_{v-1}p_{v-1}(t) - (\lambda_v + \mu_v)p_i(t) + \mu_{v+1}p_{v+1}(t)$$

$$\frac{1}{dt}p_i(t) = \lambda_{v-1}p_{v-1}(t) - (\lambda_v + \mu_v)p_i(t) + \mu_{v+1}p_{v+1}(t)$$

$$\frac{1}{dt}p_K(t) = \lambda_{K-1}p_{K-1}(t) - \mu_Kp_K(t)$$

Solving above equations by using numerical methods such as the RungeKutta method, we can investigate the stochastic behaviour of infection. Unlike deterministic models, the number of worm-infected hosts almost surely goes to zero because the state where all the worms are combated is an absorbing state. In other words, extinction of worm always occurs in the stochastic SIS model. This viewpoint provides a remarkable difference between deterministic and stochastic models.

3.2 Stochastic Kill Signal Model

Define the stochastic processes $V(t)$ and $W(t)$ which are the number of worm-infected and non-vulnerable hosts at time t , respectively. We consider the CTMC with the state $(V(t), W(t)) = (v, w)$. Then the transition rates at the state (v, w) can be described as follows:

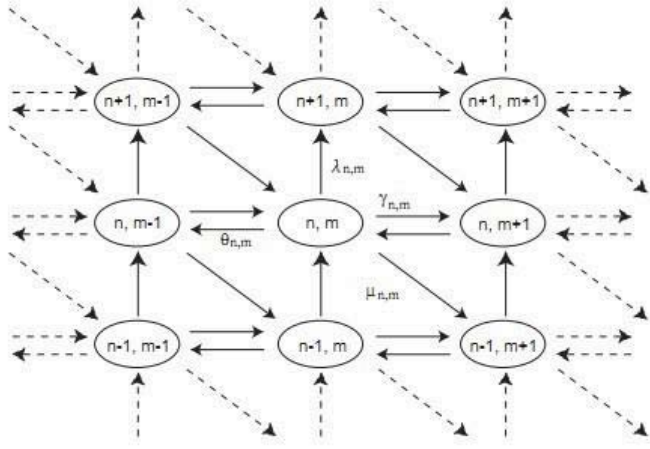


Figure 1: Transition diagram on the propagation model with KS and re-vulnerability rate

1. Worms are propagating to the other vulnerable hosts, so that the transition rate to state $(v + 1, m)$ is given by

$$\lambda_{v,w} = \beta v(K - v - w)$$

2. When worms are removed from hosts, the hosts become non-vulnerable. Hence, the transition rate to the state $(v - 1, w)$ is zero.
3. There are two cases on the removal of worm; infected hosts remove worms from themselves or worms are removed by receiving KSs. Then, the transition rate to the state $(v-1, w+1)$ is the sum of the transition rates in these two cases:

$$\mu_{v,w} = \delta v + \beta_r v w$$

4. Non-vulnerable hosts send KSs to vulnerable hosts. The transition rate to the state $(v, w + 1)$ is given by

$$\gamma_{v,w} = \beta_r w (K - v - w).$$

5. Non-vulnerable hosts can be vulnerable again by detection of unknown vulnerable factors. The transition rate to the state $(v, w - 1)$ is given by

$$\theta_{v,w} = \delta r w$$

Then the transition diagram of CTMC is depicted in Figure 1. Consider the probability:

$$p_{v,w} = \Pr\{V(t) = v, W(t) = w\}.$$

Using the above listed equations, we obtain the following differential equations:

$$\frac{1}{dt} p_{v,w}(t) = \lambda_{v-1,w} p_{v-1,w}(t) + \mu_{v+1,w-1} p_{v+1,w-1}(t) + \gamma_{v,w-1} p_{v,w-1}(t) + \theta_{v,w+1} p_{v,w+1}(t) - v_{v,w} p_{v,w}(t)$$

for $v = 0, \dots, K, w = 0, \dots, K - v$ where $v = \lambda_{v,w} + \mu_{v,w} + \gamma_{v,w} + \theta_{v,w}$. Note that $K \geq v + w$, because the total number of hosts is finite. δ -removal rate of computer virus, β_r -KS spreading rate, δ_r -re-vulnerability rate

4. Virus Propagation Model

In all the above listed models they assume that the rate of propagation is equal between any two connected nodes i.e homogeneous distribution of the rate of propagation, but the real networks deviates from this. In general most of the real networks doesn't follow this. So In our model we consider this fact and consider a $K \times K$ matrix in which an element β_{ij} indicate the rate of propagation between i th and j th node where K is the total number of nodes in the network. We take the help of stochastic KS model mentioned above for this.

1. Worms are propagating to the other vulnerable hosts, so that the transition rate to state $(v + 1, m)$ the probability that a node is infected given state (v, w) is $\frac{v}{K}$ the probability that a node is susceptible given state (v, w) is $\frac{K-v-w}{K}$ so each node can spread the virus at a rate of

$$\frac{v}{K} \sum_{j=1}^K \beta_{ij} a_{ij} (K - v - w)$$

So for all the nodes in the network the rate is given by

$$\lambda_{n,m} = \sum_{i=1}^K \frac{v}{K} \sum_{j=1}^K \beta_{ij} a_{ij} (K - v - w) = \text{sum}(I * (\beta * A * S))$$

- β -propagation matrix of computer virus ($K \times K$) β_{ij} is the i th row j th column element in β matrix.
- A-adjacency matrix of the network nodes graph ($K \times K$) a_{ij} is i th row j th column element in A matrix.
- $I - \left(\frac{v}{K} \quad \frac{v}{K} \quad \dots \quad \frac{v}{K} \right)^T_{1 \times K} S - \left(\frac{K-v-w}{K} \quad \dots \quad \frac{K-v-w}{K} \right)^T_{1 \times K}$
- K-Total number of nodes in the network
- Sum()-sum of the elements in a column vector
- $(.*)$ indicate element-wise multiplication of the two matrices of same order.

2. When worms are removed from hosts, the hosts become non-vulnerable. Hence, the transition rate to the state $(v - 1, w)$ is zero.

3. There are two cases on the removal of worm; infected hosts remove worms from themselves or worms are removed by receiving KSs. Then, the transition rate to the state $(v-1, w+1)$ is the sum of the transition rates in these two cases:

$$\mu_{v,w} = \delta v + \beta_r v w$$

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5. Non-vulnerable hosts can be vulnerable again by detection of unknown vulnerable factors. The transition rate to the state $(v, w - 1)$ is given by

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Then the transition diagram of CTMC is depicted in Figure 1. Consider the probability:

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for $v = 0, \dots, K, w = 0, \dots, K - v$ where $v = \lambda_{v,w} + \mu_{v,w} + \gamma_{v,w} + \theta_{v,w}$. Note that $K \geq v + w$, because the total number of hosts is finite. δ -removal rate of computer virus, β_r -KS spreading rate,, δ_r -re-vulnerability rate

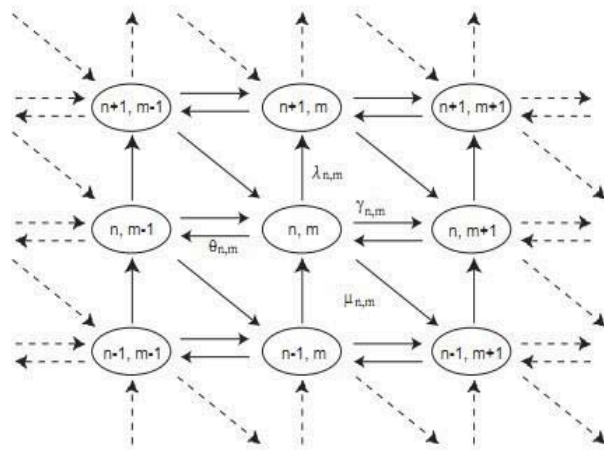


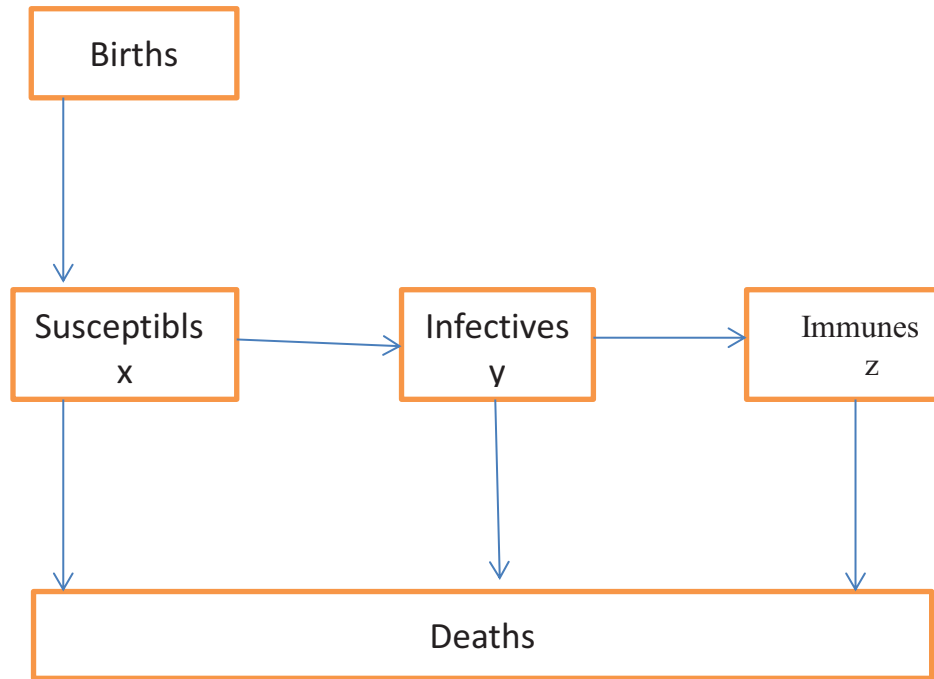
Figure-state transition diagram for the propagation model

In the above model we have considered that the rate of propagation of the virus is not same between every two nodes we considered a $K \times K$ matrix where K is the total number of nodes in the network. The elements of the matrix are function of time. One of the reasons for variable propagation rate is that the address space of a network is randomized by some randomization techniques like ASLR etc and the chance that the exploitation succeeds is a probabilistic one and takes random amount of time for the exploitation to succeed so the rate of propagation of the network worm doesnot take constant rate between every two nodes. Our model captures this fact. Most real world networks follows this fact which the other models doesnot take into consideration.

5. Deterministic SIR Model with birth and death rates

Using the case of new users connecting to the network, there is an arrival of new susceptible nodes into the population. For this type of situation birth and death rates must be included in the model. Applying the above changes to standard SIR model, the following differential equations represent our model assuming death and birth rate γ

$$\gamma(x + y + z)$$



$$\frac{dx}{dt} = -\beta xy + \gamma(y + z)$$

$$\frac{dy}{dt} = \beta xy - \gamma y - \delta y$$

$$\frac{dz}{dt} = \delta y - \gamma z$$

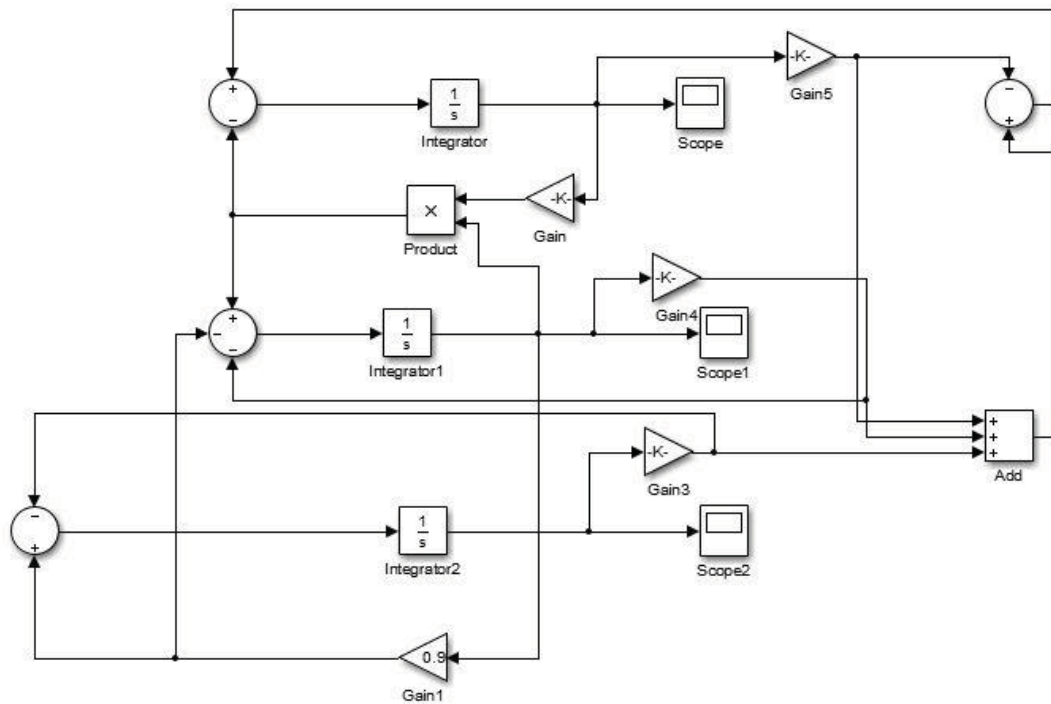
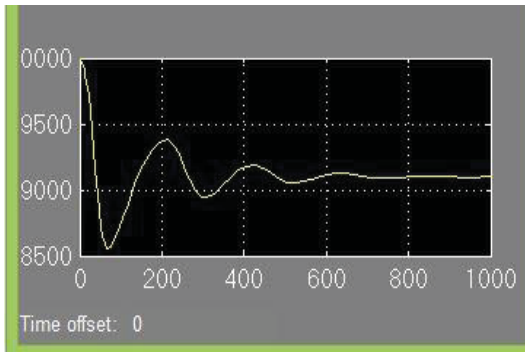
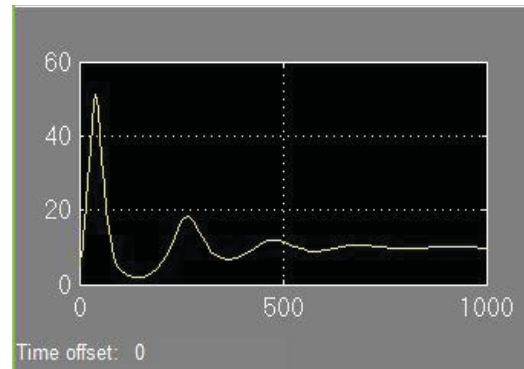


Figure-Simulink Model for the SIR model with birth and death rates

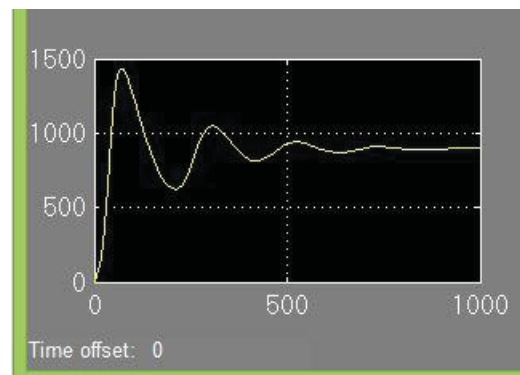
Above simulations were done using Simulink MATLAB. Following are the resulting graphs Initial assumptions are 10,000 susceptible 5 infected and 0 immune nodes $\beta = 0.0001$ $\delta = 0.01$ $\gamma = 0.9$



No. of susceptible nodes vs time(days)



No. of infected nodes vs time(days)



No. of immune nodes vs time(days)

6. Conclusion

We can use the mathematical models to measure the return on investment made by banks on security features by measuring the effectiveness of these measures in reducing financial losses in the event of a wide spread cyber attack. Using these models companies can measure the expected value of how much a company/government should spend on the security systems.

Immunization strategies must concentrate on nodes that are statistically significant. Statistically significant nodes are not necessarily limited to ones that are highly connected. The major shortcomings in existing models are they do not take into account the effect of security measures built into the system to stop the growth. As in the case of virus propagation models security protection measures

can be either deterministic or randomized. Deterministic protection measures are largely algorithm based like canary based buffer overflow protection. In such cases we agree that the time taken to overcome the protection mechanism is constant. Therefore no changes need to be made to the models proposed so far. But in case of protection mechanisms like memory layout randomization, the success of an attack is probabilistic and is subject to correctly guess the randomizing key. In such cases the reproduction factor R will be dependent on parameters such as size of randomization key. Finally, the purpose of an attack on a financial system is not to reduce productivity by taking away valuable computational resources, but to make the V2V system do a meaningful task from the attacker point of view. Hence an attack can remain latent inside the system and not be activated until the attacker chooses to launch the attack. So there is a need to evaluate the spread of an attack in a financial network by applying SEIS and SEIR models rather than simple SIS and SIR models.

In this modern era mobile banking is also becoming prominent. So we may work in the direction to build a mathematical model for virus propagation on time varying networks.

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